

I-Star index, 399

I-Star market impact model, 129, 146–148

- parameter estimation techniques, 157–162
 - guesstimate technique, 160
 - model verification, 160–162
 - non-linear optimization, 160
 - two step regression process, 157–159

I-Star model parameters, estimating, 163, 179–191

cost curves, 186

error analysis, 187–189

scientific method, 163–166

- asking question, 164
- conclusion and communication, 165–166
- data analysis, 165
- hypothesis, constructing, 164
- hypothesis, testing, 164–165
- problem, research on, 164

solution technique, 166–191

- analysis period, 176
- annualized volatility, 178
- average daily traded volume (ADV), 178
- cost, 179
- data definitions, 175–176
- first price, 178
- hypothesis, constructing, 171–173
- hypothesis, testing, 173–175
- imbalance, 176–177
- number of data points, 176
- POV rate, 179
- problem, research on, 166–171
- question, 166
- side, 177
- size, 179
- time period, 176
- turnover, 177
- universe of stocks, 176
- volume, 177
- VWAP, 178

sensitivity analysis, 181–186

statistical analysis, 187

stock specific error analysis, 189–191

K

Kolmogorov-Smirnov goodness of fit, 124–125

L

Large and small cap trading, 62

Large cap index, 398–399

Large cap stocks, 62, 167, 185

Lee & Ready tick rule, 154–155

Likelihood function, 208–209

Limit order, 23

- marketable, 23
- models, 34

Liquidation costs, 377–380

Liquidity seeking algorithms, 20

Liquidity trading, 439–440

Log price return, 194

Log returns, 196

M

Macro/micro trading goals, equations for specifying, 270–272

Macroeconomic factor models, 228–231

Macro-level strategies decision rules, 29–33

- adaptation tactic, specifying, 32–33
- implementation benchmark, selection of, 30
- optimal execution strategy, selection of, 30–32

Maker-taker model, 56

Mann-Whitney U test, 120–122

Market cost adjusted z-score, 113–114

Market expectations, 365

Market exposure, 395–396, 415

Market impact, 91–92, 301

- for a basket of stock, 243
- for a single stock order, 241–243

Market impact cost, 171–173, 377*f*, 399, 412–413, 418, 421

Market impact equation, 148–154

- comparison of approaches, 153–154
- cost allocation method, 149–151
- derivation of model, 148–149
- I* formulation, 151–153

Market impact factor scores, 384–388

- analysis, 386–388
- current state of, 386
- scores, 387*f*
- shares, derivation of, 384–386

Market impact models, 129

- definition, 129–131
- derivation of, 142
- Almgren & Chriss (AC) market impact model, 142–143
- random walk with market impact, 144–146
- random walk with price drift, 143–144

developing, 139–140

essential properties of, 140–141

graphical illustrations of, 131–139

- market impact price trajectories, 138–139
- price trajectory, 131–132
- supply-demand equilibrium, 132–135
- temporary decay formulation, 137–138
- temporary impact decay function, 135–137

I-Star market impact model, 146

model formulation, 147–156

- I-Star, 147–148
- market impact equation, 148–154
- underlying data set, 154–156

parameter estimation techniques, 157–162

- guesstimate technique, 160
- model verification, 160–162
- non-linear optimization, 160
- two step regression process, 157–159

permanent impact, 130–131

research step for, 166–171

temporary impact, 130

Market impact price trajectories, 138–139

Market making (MM), 14–15

Market microstructure, 47

- empirical evidence, 61–76
 - day of week effect, 65–67
 - intraday trading profiles, 67–73
 - intraday trading stability, coefficient of variation, 72–73
 - special event days, 73–76
 - trading volumes, 61–63
 - volume distribution statistics, 63–65

equity exchanges, 57

flash crash, 76–85

- comparison with previous crashes, 84–85
- empirical evidence from, 79–83

literature, 49–51

NASDAQ select market maker program, 60–61

new market structure, 51–56

new NYSE trading models, 57–60

- designated market makers (DMM), 58–59
- supplemental liquidity providers (SLPs), 59–60
- trading floor brokers, 60

order priority, 57